

An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments

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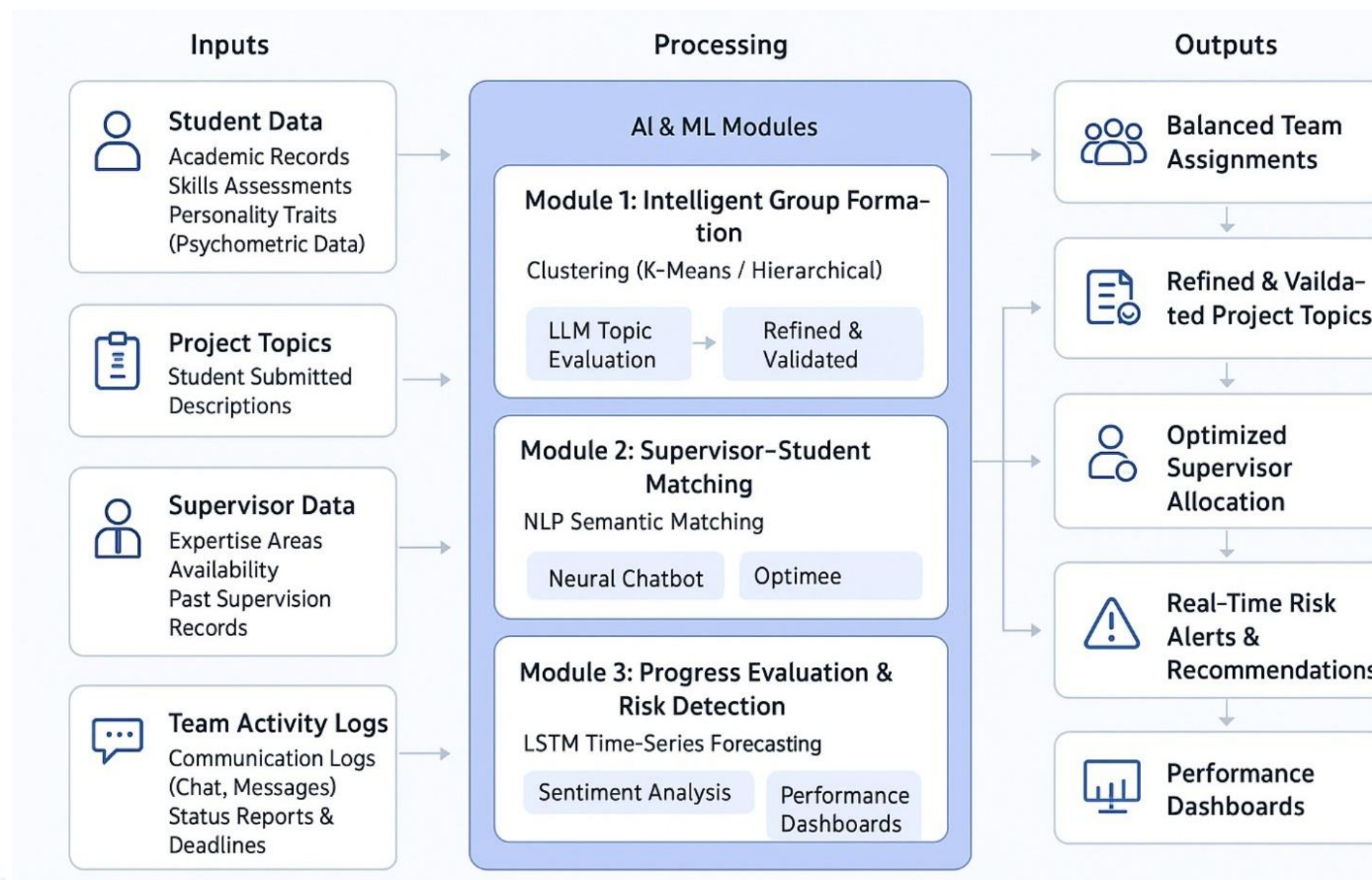
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INTELLIGENT GROUP FORMATION AND TOPIC EVALUATION

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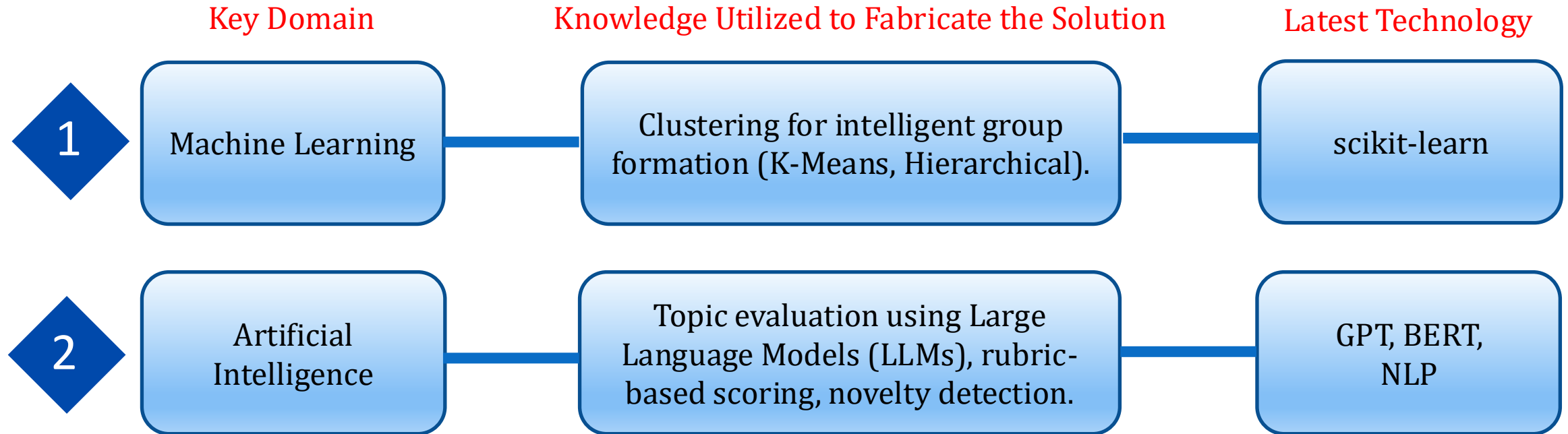
Intelligent Group Formation And Topic Evaluation

Research Gap

- Mainly rely on academic or demographic factors, ignoring collaboration style and soft skills.
- Limited novelty detection and fail to provide actionable feedback for improving topics.

Existing system	Shortcomings	Proposed Solution
GroupEng	Rule based(instructors need enter factors they desire)	Using K-Means clustering(based on similarity)
TEAMMAKER	Mainly rely on demographic data , risk of weaker groups	Multi attribute inputs(GPA, Skills, Interests, Traits)
Du (2024) – Auto-evaluation of Reports with LLMs	Uses LLMs to provide feedback on final student reports	Early-stage topic evaluation with LLMs to refine ideas before project starts
Manual Process	Time consuming, inconsistent novelty detection	LLMs Provide structured consistent, rubric based topic evaluation Project-focused allocation + mentorship

Intelligent Group Formation And Topic Evaluation



Intelligent Group Formation And Topic Evaluation

Evaluation

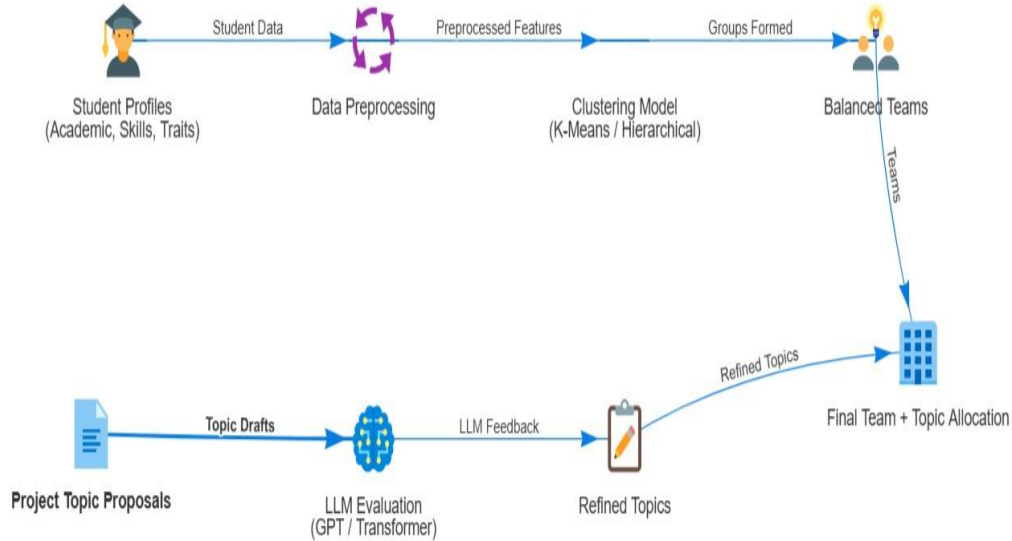
- Form group with our system, existing system compare GPA balance, skill diversity, fairness index, satisfaction survey.
- Higher satisfaction rate (> 70% positive feedback)
- Compare our LLM evaluation vs existing methods (metrics agreement with supervisor scores, novelty detection accuracy, usefulness of feedback)
- Highest satisfaction rate (≥ 80 % with supervisors)

Data Availability

- Student Data(GPA, Grades, technical & soft skills)
- Project topics(Current students proposals, Past project topics)
- All students data stored without personal identifiers, follow university policies)

Intelligent Group Formation And Topic Evaluation

System Overview



Real-world scenario

- Students submit their profiles: GPA, grades, skills, interests, and draft project topic proposals via form.
- **ML clustering** (K-Means/Hierarchical) automatically creates **balanced groups**.
- **LLM-based evaluation** checks topics for clarity, novelty, feasibility, and relevance.
- Students refine topics based on automated feedback.

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SUPERVISOR-STUDENT MATCHING AND NLP- BASED PROJECT GUIDANCE SYSTEM

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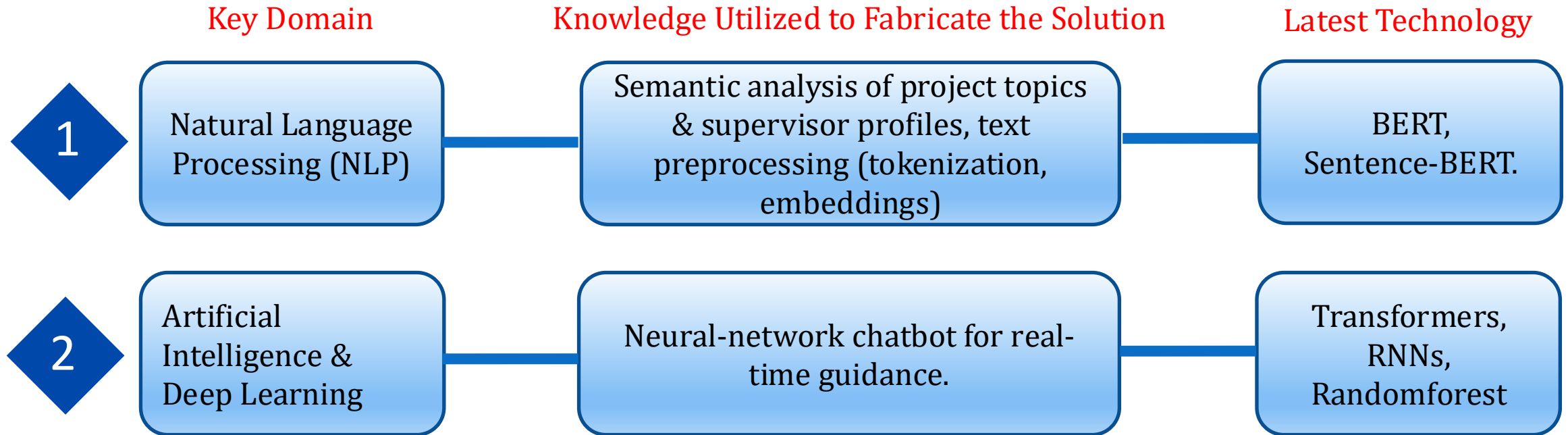
Supervisor-Student Matching and NLP- Based Project Guidance System

Research Gap

- Supervisor allocation is manual/keyword-based → mismatches.
- Guidance is fragmented (emails, meetings) → delays & poor guidance.

Existing system	Shortcomings	Proposed Solution
FYPMS	Manual/keyword allocation	NLP-based semantic matching + guidance
EasyChair	Rules-based, no mentoring	Interest-based + chatbot support
Teams / Moodle	Communication only, no allocation	Combines allocation + guidance
Brightspace	Coursework-focused, not projects	Project-focused allocation + guidance

Supervisor-Student Matching and NLP- Based Project Guidance System



Supervisor-Student Matching and NLP- Based Project Guidance System

Evaluation

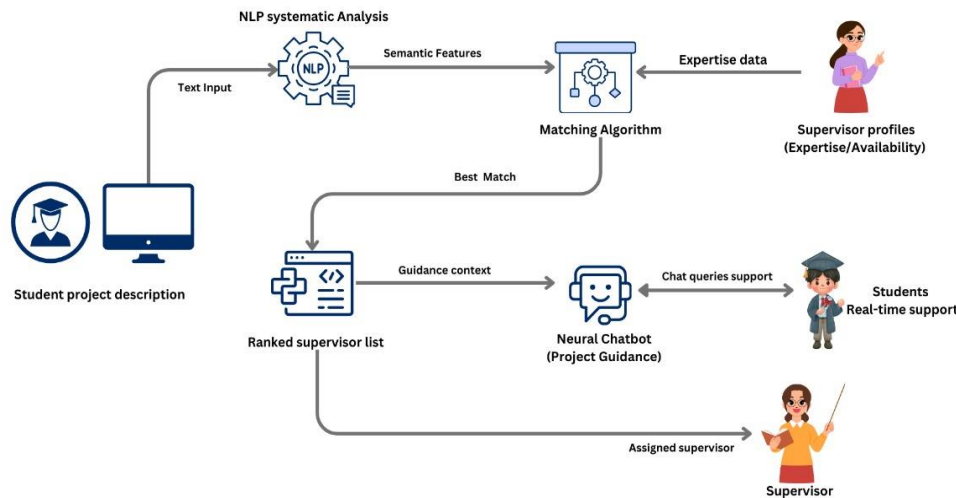
- Supervisor Matching Accuracy
- Chatbot Performance
- User feedback and Satisfaction

Data Availability

- Student Project Topics (Current project proposals + past project records)
- Supervisor Profiles (expertise, workload capacity, availability)
- All Student and Supervisor information stored securely, without personal identifiers, following university policies.

Supervisor-Student Matching and NLP- Based Project Guidance System

System Overview



Real-world scenario

- Students submit project topics.
- System suggests the best supervisor based on expertise.
- Chatbot handles routine queries (deadlines, formatting, process).

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AI-DRIVEN PROGRESS EVALUATION & RISK DETECTION

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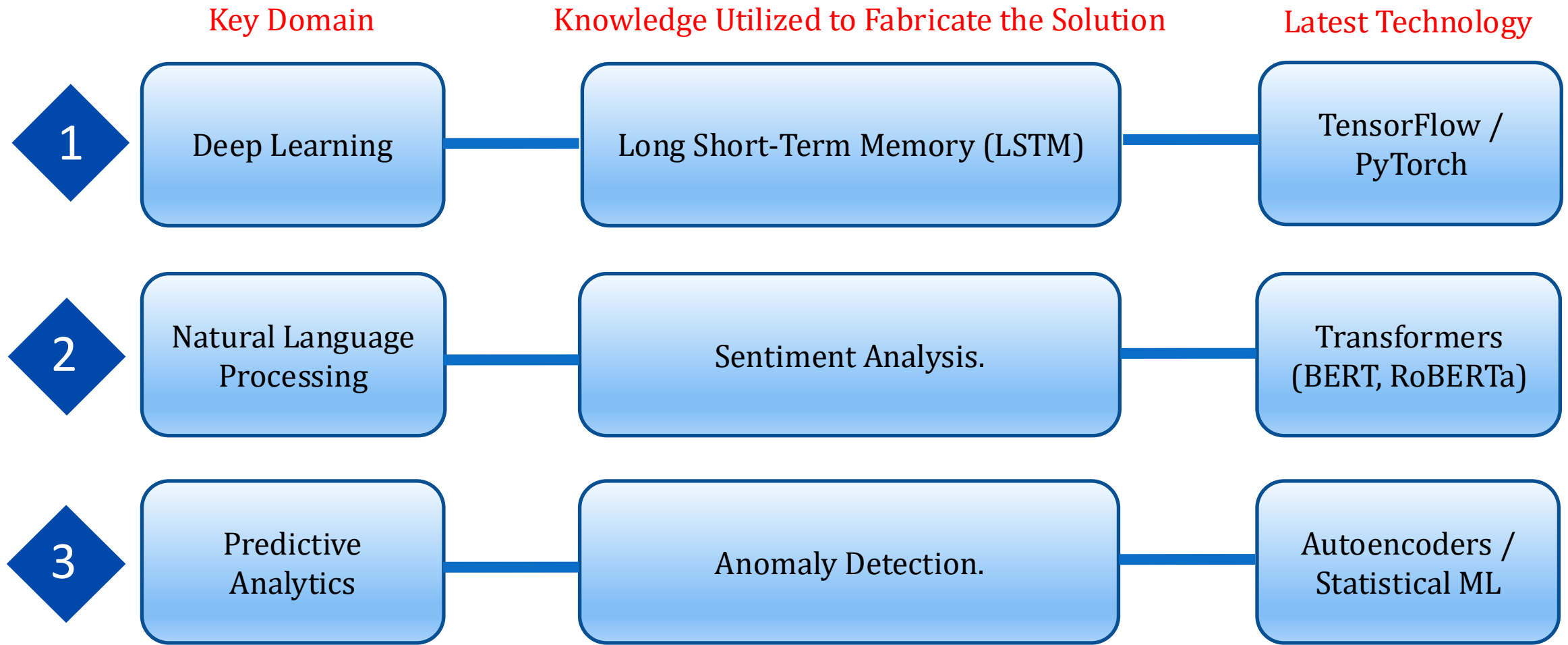
AI-Driven Progress Evaluation & Risk Detection

Research Gap

- Hidden issues remain undetected until late stages.
- No integrated framework for a proactive project monitoring and early risk alerts.

Existing system	Shortcomings	Proposed Solution
<u>SpringerOpen</u>	No predictive or proactive monitoring	Integrates predictive LSTM forecasting to provide early warnings rather than just past data summaries
<u>SpringerLink</u>	Analytical methods used separately	Combines sentiment analysis with forecasting and anomaly detection for comprehensive, integrated risk evaluation.
<u>MDPI</u>	Absence of real-time dashboards and early intervention tools	Provides real-time dashboards with live risk alerts and progress insights for proactive interventions.

AI-Driven Progress Evaluation & Risk Detection



AI-Driven Progress Evaluation & Risk Detection

Evaluation

- Forecasting Accuracy: RMSE, MAPE for LSTM predictions
- Sentiment Analysis: Precision, Recall, F1-score
- Anomaly Detection: Detection Rate vs False Alarms
- Usability Testing: Feedback from supervisors and students

Data Availability

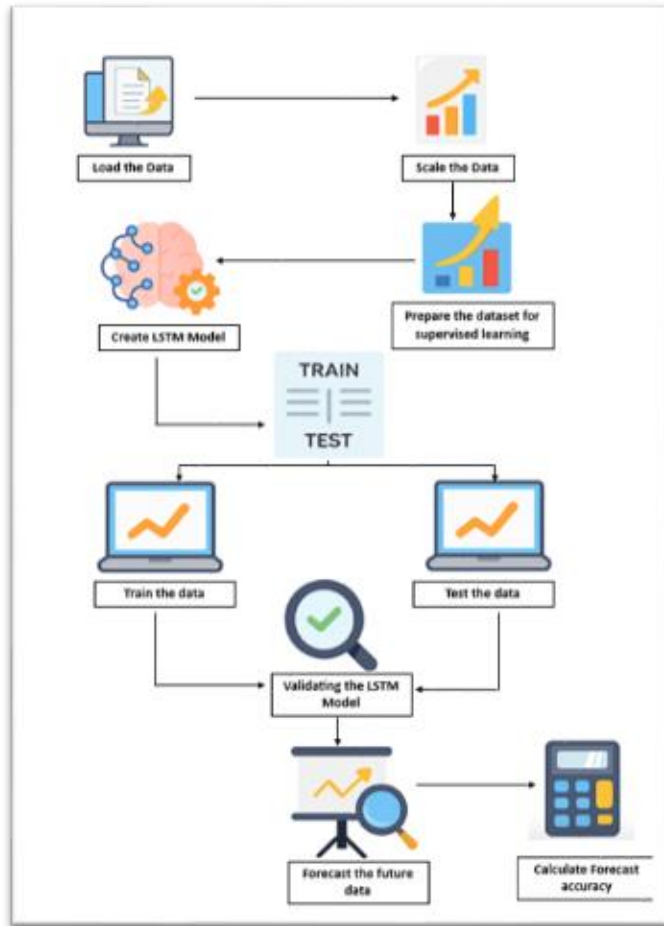
- Student Data - GPA / Grades, Attendance, Submission history, Age, Gender
- Project Data - Project logs, Submission history, task schedule, progress reports
- Communication Data- Supervisor feedback comments, Team chat logs. Meeting transcripts.

Ethical Clearance

- Protecting student privacy
- Ensuring no personal data misuse
- Role-based access control

AI-Driven Progress Evaluation & Risk Detection

System Overview



Real-world scenario

- Students and supervisors upload project updates, logs, and communications into the system.
- LSTM-based forecasting predicts whether project milestones are on track or at risk of delay.
- Sentiment analysis on team communications highlights issues such as disengagement, conflicts, or low morale.
- Anomaly detection flags irregular activity patterns. Supervisors and students receive real-time dashboards and alerts, enabling timely interventions to prevent project failures.

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Target Audience

Universities and Higher Educational Institutes.
Research Institutes and Academic Councils.

Pricing Strategy

Tiered Subscription plan
Discounted academic rates for developing regions to encourage adoption.

Marketing & Sales

Direct outreach to the universities, colleges.
Collaborations with education boards.